Greenfield School Results Project

Excel Student Guide: clean the data, find the answers, advise the principal

Welcome! This guide shows you exactly what to do, one step at a time. Read it slowly and follow the steps in order. You can do this.

1. What is this project about?

You work as the data assistant at **Greenfield Secondary School**. The school has **240 students** in **6 classes** (JSS1A, JSS1B, JSS2A, JSS2B, JSS3A, JSS3B). The results of all three terms are in one Excel file. The principal wants answers to 4 questions:

- ✓ **Which class is doing best, and which class needs the most help?**
- ✓ **Which subject is the hardest for students?**
- ✓ **Are students improving from First Term to Third Term?**
- ✓ **Which students need extra support?**

But there is a problem. The data was typed by different teachers, so it is messy. Some class names are written in different ways, some students appear twice, some scores are missing or typed wrongly.

Your job: clean the data, use formulas to find the answers, build a one-page summary, and write a short message to the principal.

MEET THE DATA

Open **Greenfield_School_Results_Dataset.xlsx**. There is one sheet called **Results**. One row = one student in one term (so each student has 3 rows).

Column	What it means
Student ID	The number of the student, like STU001. It never changes.
Student Name	The full name of the student
Gender	Male or Female
Class	JSS1A, JSS1B, JSS2A, JSS2B, JSS3A or JSS3B
Term	First Term, Second Term or Third Term
Maths, English, Basic Science, Social Studies, Computer Studies, Civic Education	The score for each subject, from 0 to 100

THE RULES OF THE SCHOOL (VERY IMPORTANT)

Rule	Details
Pass mark	A student passes a term if the **average** of the subjects is **50 or more**.
Grade scale	A = 70 and above, B = 60 to 69, C = 50 to 59, D = 45 to 49, E = 40 to 44, F = below 40. Use the exact average. Do not round the Average cell. Only change how it looks (1 decimal place).
Blank score	The student did not write that subject. **Leave it blank. Do not change it to 0.** Total and Average will skip blank cells.
Score above 100	This is a typing mistake: an extra zero was added (for example 780). Remove the extra zero (780 becomes 78).
Same student twice	A student can appear only once in each term. If Student ID and Term appear twice, delete the second copy.

WHAT YOU MUST DO

There are 8 tasks. Do them in order. Use one Excel file. Keep the original **Results** sheet untouched and do your work in new sheets, named like this: **Clean Data**, **Class Summary**, **Subjects**, **Improvement**, **At Risk**, **Summary**.

TASK 1. CLEAN THE DATA (CLEAN DATA SHEET)

Copy the Results sheet into a new sheet called **Clean Data** and fix all the mistakes. Then write a short **Cleaning Report** (on any sheet): what was wrong, how many, and how you fixed it.

- Fix the class names so only **6 classes** exist. Fix the names (extra spaces and CAPITAL letters). Make Gender only **Male** and **Female**.
- Some scores are stored as text (they sit on the left of the cell). Change them to real numbers. Fix the scoresabove 100.
- Clean first, **then** remove the duplicates. Some copies are written differently, so Remove Duplicates will not find them until the data is clean.
- **Check yourself:** after cleaning you should have **720 rows**, **240 students** and **6 classes**.

TASK 2. TOTAL, AVERAGE AND GRADE

Add three columns: **Total**, **Average** and **Grade** for every row, using formulas (not typed numbers).

- Hint: SUM, AVERAGE and IFS (or nested IF). Look at the grade scale in the rules table.

TASK 3. RANK THE STUDENTS

Add a **Class Rank** column: the position of the student in his or her class, for that term. Then list the **Top 10 students in the Third Term** (name, class, average) and the **top 3 in each class** in the Third Term.

- Hint: sort the Third Term rows by Average, or use this formula in plain words: count how many students in the same class and term have a higher average, then add 1 (see the help sheet).

TASK 4. CLASS SUMMARY (CLASS SUMMARY SHEET)

For each class and each term, show the **number of students**, the **average score** and the **pass rate** (the percentage of students with an average of 50 or more). Which class is best in the Third Term?

- Hint: AVERAGEIFS and COUNTIFS. A PivotTable is also fine for the average.

TASK 5. WHICH SUBJECT IS THE HARDEST? (SUBJECTS SHEET)

Find the **average score of each subject** (all terms together) and also for each term. Name the hardest subject and the easiest subject.

- Hint: AVERAGE on each subject column of your clean data.

TASK 6. ARE STUDENTS IMPROVING? (IMPROVEMENT SHEET)

For every student, compare the Third Term average with the First Term average. Show the change, and mark each student **Improved** or **Dropped**. How many students improved? Who are the **5 most improved** students?

- Hint: AVERAGEIFS with Student ID and Term, then Third minus First. Use RANK or LARGE to find the top 5.

TASK 7. STUDENTS WHO NEED SUPPORT (AT RISK SHEET)

List all students whose **Third Term average is below 50**. Show how many they are. Colour these averages red with **conditional formatting**.

- Hint: filter your Clean Data sheet, or use COUNTIFS for the number.

TASK 8. SUMMARY SHEET AND MESSAGE TO THE PRINCIPAL

Build one clean **Summary** sheet with: (1) five key numbers (for example total students, Third Term pass rate, hardest subject, best class, number of students below 50), (2) **three charts**: class average by term, average score of each subject, and pass rate by class in the Third Term, and (3) a **message to the principal**.

- **Charts**: choose the right type (columns, bars or lines), give every chart a **clear title** and **axis titles**, keep the colours simple, and remove anything that is not needed.
- **Write 3 to 5 simple sentences. Answer**: Which class needs the most help? Which subject should the schoolfocus on? How many students need support? What is one thing you would like to know before the school makes a big decision?

BONUS QUESTIONS (EXTRA MARKS)

B1. Which class improved the most from First Term to Third Term (class average, Third minus First)?

B2. Add a PivotTable or chart that shows the average score for each subject in each class.

B3. Make your summary interactive: build a PivotTable and PivotChart with **slicers** for Class and Term, so the principal can click and filter.

EXCEL HELP SHEET

I want to...	I type...
Remove extra spaces	=TRIM(B2)
Fix CAPITAL letters in a name	=PROPER(TRIM(B2))
Change text to a number	=VALUE(F2) (or use Text to Columns)
Remove a space inside text	=SUBSTITUTE(D2," ","")
Make letters CAPITAL	=UPPER(D2)
Add the scores in a row	=SUM(F2:K2)
Find the average (blank cells are skipped)	=AVERAGE(F2:K2)
Give a grade	=IFS(M2>=70,"A",M2>=60,"B",M2>=50,"C",M2>=45,"D",M2>=40,"E",TRUE,"F")
I want to...	**I type...**
Average of a group	=AVERAGEIFS(M:M, D:D, "JSS1A", E:E, "Third Term")
Count rows that match	=COUNTIFS(D:D, "JSS1A", E:E, "Third Term", M:M, ">=50")
Rank inside a class and term	=SUMPRODUCT((D$2:D$721=D2)*(E$2:E$721=E2)*(M$2:M$721>M2))+1
Find the biggest or smallest	=MAX(range) or =MIN(range) or =LARGE(range, 2)
Find the name that matches	=INDEX(B:B, MATCH(A2, A:A, 0))
Colour cells under 50	Home > Conditional Formatting > Highlight Cells Rules > Less Than

Make a chart	Select the data, then Insert > pick a chart. Add a title and axis titles: Chart Design > Add Chart Element
Add a slicer	Click inside your PivotTable, then Insert > Slicer

SMALL TIPS

- A formula always starts with **=**.
- Use **$** to lock a range, for example $D$2:$D$721, so it does not move when you copy the formula down.
- If Excel says #NAME?, check the spelling of the formula. If you have an older Excel without IFS, usenested IF.
- Do not type numbers that a formula can calculate. Marks are given for formulas.
- Save your work often and keep a backup copy.

WHAT TO HAND IN

- Your Excel file named **results_yourname.xlsx** (for example results_amaka.xlsx), with all your sheets.
- A screenshot of your Summary sheet.
- Your message to the principal (3 to 5 sentences, also typed in the Summary sheet).

Where do I send it? I will tell you in class where to send your work. Please do not send it before I confirm the place.

HOW YOU WILL BE MARKED

What	Marks
Correct answers (Tasks 2 to 7)	40
Data cleaning and cleaning report (Task 1)	20
Neat workbook: formulas (not typed numbers), clear sheet names, good formatting	15
Summary sheet, 3 charts (right type, title, axis titles, easy to read) and message to the principal (Task 8)	25
Bonus (B1, B2 and B3)	up to 5 extra

IF SOMETHING GOES WRONG

Problem	What to do
My numbers are different from my friend's	Did you clean the data first? Did you remove the duplicates only after cleaning? Did you leave blank scores blank?
My average looks wrong	Check that all scores are numbers (not text). Text scores are skipped by AVERAGE.
Excel shows #NAME? or #VALUE!	Check the spelling of the formula, the brackets and the commas.
I am stuck	Ask me in class, or ask a classmate. Show your formula and the error message.

Good luck! Take it one step at a time. Every expert started as a beginner.